Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Randomized Parallel-group, Placebo-controlled, Double-blind, Event-driven, Multi-center Phase 2 Clinical Outcome Trial of Prevention of Arteriovenous Graft Thrombosis and Safety of MK-2060 in Patients With End Stage Renal Disease Receiving Hemodialysis **Short Title/Acronym:** FXI Hemodialysis Study (MK-2060-007) **Protocol Number:** MK-2060-007 **NCT Number:** NCT05027074 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-2060 (Anti-Factor XI Monoclonal Antibody) **Phase of Development:** Phase 2 **Medical Monitor:** Merck Sharp & Dohme Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of two different doses of MK-2060 (a monoclonal antibody against Factor XI) compared to placebo in increasing the time to first occurrence of an arteriovenous graft (AVG) thrombosis event in participants with End Stage Renal Disease (ESRD) receiving hemodialysis. **Secondary Objectives:** To aid in dose selection for future studies by evaluating safety profiles—specifically major bleeding and clinically relevant non-major bleeding—as well as analyzing pharmacokinetics and pharmacodynamics.

2.2 Background & Rationale Patients with ESRD receiving hemodialysis via an AVG frequently face complications with graft occlusion due to thrombosis. MK-2060 is an investigational monoclonal antibody that targets the contact pathway by inhibiting Factor XI, which prevents downstream blood coagulation without broadly disabling the hemostatic system. The trial seeks to establish whether this novel mechanism can safely reduce the risk of major thrombotic cardiovascular events (like AVG thrombosis) without disproportionately increasing the risk of bleeding compared to placebo.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Quadruple Blind) **Randomization:** Randomized (Parallel-group)

3.2 Planned Enrollment & Duration Target N: Approximately 489 evaluable patients **Number of Sites:** 120 locations globally

Estimated Study Start: 17-Sep-2021 **Estimated Primary Completion:** 30-Oct-2024

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Current diagnosis of End Stage Renal Disease (ESRD). 2. Receiving hemodialysis (including hemodiafiltration) ≥ 3 times per week for a minimum of 3 hours per session via a mature, normally functioning, uninfected AVG, with at least 75% of the sessions meeting these criteria over the 4 weeks prior to randomization. 3. Female participants must not be pregnant or breastfeeding, must not be a woman of child-bearing potential (WOCBP), or if WOCBP, must agree to follow contraceptive guidance during the intervention and for at least 90 days after the last dose.

4.2 Exclusion Criteria 1. Recent history of cancer (< 1 year). Non-melanoma skin cancers are allowed. 2. Mechanical/prosthetic heart valve, recent hemorrhagic or lacunar stroke (< 1 month), or recent evidence of bleeding requiring hospitalization/medical attention. 3. Currently receiving or planning to receive anticoagulants or antiplatelet medications (intradialytic heparin and aspirin are permitted). 4. Planning to receive a living donor renal transplant or arteriovenous fistula (AVF) placement within 12 months.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-2060 20 mg (loading dose QOD week 1, then QW) OR MK-2060 6 mg (loading dose QOD week 1, then QW), versus Placebo (normal saline). **Route of Administration:** Intravenous (IV) infusion during dialysis. **Duration of Treatment:** Event-driven clinical outcome timeline.

5.2 Concomitant & Prohibited Therapy Permitted: Intradialytic heparin and aspirin. **Prohibited:** Concurrent administration of other systemic anticoagulants or antiplatelet medications.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Confirmation of AVG function
Baseline	Day 0	Randomization, First Dose (Loading dose 1)
Interim	Weeks 1+	IV Administration QOD during Week 1 (3 doses), then QW during scheduled hemodialysis; Safety Labs, Efficacy Assessment
End of Study	Event-Driven	Final Vital Signs, Exit Interview, Follow-up 90 days post-last dose

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to first occurrence of AVG thrombosis event (defined as sudden occlusion of the AVG requiring thrombectomy/thrombolysis, or clinical evidence of thrombosis with surgical, radiological, or pathological confirmation). **Measurement Method:** Evaluated by a blinded independent Clinical Adjudication Committee (CAC) utilizing a time-to-event methodology.

7.2 Secondary & Exploratory Endpoints * Incidence of Major Bleeding Events (e.g., fatal bleeding, symptomatic bleeding in a critical area, drop in hemoglobin ≥ 20 g/L). * Incidence of Clinically Relevant Non-Major Bleeding Events requiring medical intervention. * Number of participants discontinuing study intervention due to an Adverse Event (AE).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Untoward medical occurrences collected continuously, temporally associated with the use of study intervention whether or not considered related to the MK-2060. **Serious Adverse Events (SAEs):** Reported promptly, with strict observation for bleeding complications or hemorrhage. **Data Monitoring Committee (DMC):** Blinded independent Clinical Adjudication Committee (CAC) established to verify AVG thrombosis events and review major bleeding events.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE/bleeding evaluations. **Sample Size Justification:** Utilizes an event-driven methodology to power the study sufficiently to detect a superior time-to-event increase in the MK-2060 groups versus placebo. **Handling of Missing Data:** Handled per standard time-to-event (e.g., Kaplan-Meier/Cox proportional hazards models) right-censoring methods.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous monitoring across the 120 global sites, tracking IV infusion adherence during hemodialysis. **Audit Trail:** Robust centralized data lock processing with adjudication tracking for all sudden occlusion and thrombectomy endpoints.

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11. Study Identifiers & Governance

Primary ID: MK-2060-007 **Registry Number:** NCT05027074 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** FXI Hemodialysis Study **Official Regulatory Title:** A Randomized Parallel-group, Placebo-controlled, Double-blind, Event-driven, Multi-center Phase 2 Clinical Outcome Trial of Prevention of Arteriovenous Graft Thrombosis and Safety of MK-2060 in Patients With End Stage Renal Disease Receiving Hemodialysis

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12. Clinical Framework (Study Specific)

Primary Condition: End Stage Renal Disease (ESRD) **Diagnosis Threshold:** Hemodialysis via arteriovenous graft (AVG) ≥ 3 times per week **Medical Methodology:** Investigational Anticoagulant Therapy (Monoclonal Antibody) **Optimization Target:** Prevention of Arteriovenous Graft Thrombosis

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13. Study Procedures & Methodology

Management Pathway: Hemodialysis pathway and vascular access management **Education Protocol (HCP):** IV infusion administration protocols during hemodialysis **Education Protocol (Patient):** Bleeding risk and AVG occlusion symptom awareness **Quality Audit Mechanism:** Blinded independent Clinical Adjudication Committee (CAC) assessment

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14. Observation & Follow-Up Schedule

Total Duration: Event-driven timeline (plus 90 days follow-up after the last dose) **Onsite Visit Cadence:** Every other day (QOD) during Week 1, then weekly (QW) during scheduled hemodialysis **Remote Monitoring Frequency:** Regular AE monitoring intervals **Checklist Review Interval:** Continuous during scheduled hemodialysis sessions

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] |** ___ **|** ***[DD-MMM-YYYY]*** **| |** ***Lead Statistician*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |